

From Regulation to Rule Pack

How we turn the law into machine-checkable JSON — and keep every rule tied to its exact legal citation.

1 Read the regulation	2 Pick label-verifiable provisions	3 Design one rule schema	4 Encode clause by clause, keep the citation	5 Validate with automated tests
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EXAMPLE 1 Maximum Retail Price

THE PROVISION

Rule 6(1)(e) - LMPC 2011

Every retail package must declare the **Maximum Retail Price**. It must be printed as 'Maximum Retail Price Rs ____ **inclusive of all taxes**' and shown on the **principal display panel**.



THE ENCODED RULE

legal_metrology_2011.json

```
{
  "id": "mrp",
  "label": "Maximum Retail Price (MRP)",
  "legal_reference": "Rule 6(1)(e)",
  "condition": "always",
  "required": true,
  "severity": "critical",
  "checks": [
    { "type": "presence" },
    { "type": "format",
      "regex": "maximum retail price" },
    { "type": "format",
      "regex": "inclusive .. all taxes" },
    { "type": "value",
      "validator": "positive_currency" },
    { "type": "placement",
      "panel": "principal_display_panel" }
  ]
}
```

JSON trimmed for display (messages & full regex omitted).

EXAMPLE 2 Minimum letter height of the net-quantity declaration

THE PROVISION

Rule 7(2), Table-I

The net-quantity numerals must be at least a minimum height, set by the area of the **principal display panel**:

Principal display panel area	Min. height
up to 50 cm ²	1.0 mm
50 to 100 cm ²	1.5 mm
100 to 500 cm ²	2.5 mm
500 to 2500 cm ²	4.0 mm
over 2500 cm ²	6.0 mm



THE ENCODED RULE

legal_metrology_2011.json

```
"font_height_table": {
  "legal_reference":
    "Rule 7(2), Table-I",
  "thresholds": [
    { "max_area_cm2": 50,
      "min_height_mm": 1.0 },
    { "max_area_cm2": 100,
      "min_height_mm": 1.5 },
    { "max_area_cm2": 500,
      "min_height_mm": 2.5 },
    { "max_area_cm2": 2500,
      "min_height_mm": 4.0 },
    { "max_area_cm2": null,
      "min_height_mm": 6.0 }
  ]
}
```

JSON trimmed for display (messages & full regex omitted).

Why encode it by hand instead of auto-scraping the PDF?

Legal text carries clause numbers, provisos and conditions that generic PDF extraction gets wrong — and a single incorrect citation makes an inspector distrust the whole tool. So each provision is read and translated into the schema by hand, and every rule stores its exact `legal_reference`. Any verdict the engine returns therefore traces straight back to the rule in law. Clause numbers and the Table-I values are taken verbatim from the 2011 Rules as amended (G.S.R. 629(E), 23 June 2017); exact gazette wording is verified at indiacode.nic.in.